

# Challenges, Ethics, and Future Implications

## Introduction to Large Language Models for Data Engineers

Data Engineering Boot Camp

July 01, 2025

# Agenda

Learning Outcomes

Technical and Operational Challenges

Data Quality and Ethical Considerations

Future Implications for Data Engineers

Q&A and Discussion

Series Conclusion

# Learning Outcomes

By the end of this lesson, you will be able to:

- ▶ Identify key technical challenges in LLM deployment
- ▶ Analyze ethical considerations in data privacy and bias
- ▶ Evaluate career implications for data engineering roles
- ▶ Discuss governance frameworks for responsible LLM deployment

# Computational Resource Constraints

## Memory Requirements:

- ▶ 7B parameters: 14GB RAM (FP16)
- ▶ 30B parameters: 60GB RAM
- ▶ 70B+ parameters: 140GB+ RAM

## Cost Implications:

- ▶ High-end GPU: \$15K–30K/unit
- ▶ Inference Server: \$100K+
- ▶ Training: Millions of dollars

**Analogy:** Like early Hadoop clusters, but starting at enterprise scale.

# Infrastructure Scaling Challenges

## Database Scaling

- ▶ Read replicas, sharding, caching
- ▶ Standard servers
- ▶ Linear scaling

## LLM Scaling

- ▶ Model parallelism, memory optimization
- ▶ Specialized hardware
- ▶ Complex compute interdependencies

# Latency and Throughput Limitations

## **Latency:**

- ▶ Database: 1–100ms
- ▶ Web API: 100–500ms
- ▶ LLM Inference: 1–30s

## **Throughput (Single GPU):**

- ▶ 10–50 concurrent users
- ▶ 20–100 tokens/second
- ▶ 2K–32K token context

**Solutions:** Load balancing, batching, caching

# Black-Box Nature and Debugging

## Traditional Debugging

- ▶ Error → Stack Trace → Fix

## Strategies:

- ▶ Input-output logging
- ▶ Regression/A/B testing
- ▶ Automated quality metrics

## LLM Debugging

- ▶ Wrong Output → Prompt Engineering
- ▶ No stack traces, no visibility

# Model Versioning and Deployment

## **Versioning Challenges:**

- ▶ Models: 10–100GB binary files
- ▶ No meaningful diffs
- ▶ Opaque changes

## **Deployment Complexity:**

- ▶ Model/Data drift
- ▶ Hardware dependencies
- ▶ Complex rollbacks



# Bias in Training Data

## Sources:

- ▶ Internet: Societal biases
- ▶ Historical data: Outdated views
- ▶ Curated datasets: Selection bias

## Types:

- ▶ Representation: Underrepresented groups
- ▶ Confirmation: Reinforces stereotypes
- ▶ Cultural: Western/English-centric

# Data Privacy and Compliance

## **Challenges:**

- ▶ Training data in model weights
- ▶ Memorization risks
- ▶ Hard to delete data

## **Compliance:**

- ▶ GDPR: Retraining for data deletion
- ▶ Residency: On-premises deployment
- ▶ Audit: Limited explainability

# Hallucinations and Reliability

**Hallucinations:** Plausible but incorrect outputs **Why They Occur:**

- ▶ Optimize for plausibility
- ▶ No real-time data
- ▶ Always generate responses

**Mitigation:**

- ▶ Retrieval-Augmented Generation
- ▶ Confidence scoring
- ▶ Multi-model validation

# Governance Frameworks

## **AI Governance:**

- ▶ **Data:** Audit sources, validate inputs/outputs
- ▶ **Model:** Versioning, testing, monitoring
- ▶ **Usage:** Access controls, human oversight

## **Best Practices:**

- ▶ Input/output validation
- ▶ Performance/quality monitoring
- ▶ Documentation and auditability

# Emerging Efficiency Improvements

## **Technical Advances (1-2 years):**

- ▶ Quantization, distillation, pruning
- ▶ Specialized chips, edge deployment
- ▶ 10x cost reductions

## **Infrastructure:**

- ▶ Current: ML expertise needed
- ▶ Future: LLM deployment like databases

# New Roles: MLOps and AI Engineering

## **MLOps Engineer:**

- ▶ Model deployment, monitoring, resource management
- ▶ Skills: Pipeline orchestration, IaC, observability

## **AI Engineer:**

- ▶ Prompt engineering, model integration, QA
- ▶ Skills: Data flows, API design, testing

# Evolution of Data Engineering

## **Traditional Focus:**

- ▶ Data Sources → ETL → Warehouse → Analytics

## **AI-Enhanced Focus:**

- ▶ AI training data, model operations
- ▶ NL interfaces, automated insights
- ▶ Intelligent monitoring

# Required Skills

## **Core Skills:**

- ▶ Data modeling, pipeline engineering
- ▶ System design, performance optimization

## **New Skills:**

- ▶ AI/ML concepts, prompt engineering
- ▶ Bias detection, explainability
- ▶ AI governance and compliance



# Discussion Questions

1. Which technical challenges will impact your work most?
2. How would you implement bias detection in LLMs?
3. What steps will you take to prepare for AI integration?
4. What governance frameworks are critical for your organization?

# Key Takeaways

- ▶ Technical challenges follow familiar patterns
- ▶ Ethical considerations need systematic approaches
- ▶ Career evolution offers opportunities for skill growth
- ▶ AI governance builds on data governance practices

# Series Recap

- ▶ **Lesson 1:** LLM foundations and scale
- ▶ **Lesson 2:** Architecture and deployment
- ▶ **Lesson 3:** Challenges, ethics, career implications

## Next Steps:

- ▶ Experiment with LLM APIs
- ▶ Learn ML fundamentals
- ▶ Network with AI practitioners

# Additional Resources

- ▶ **Getting Started:**
  - ▶ OpenAI API, Hugging Face, Google Colab
- ▶ **Professional Development:**
  - ▶ Coursera/edX ML courses
  - ▶ MLOps community, AI ethics frameworks
- ▶ **Industry Insights:**
  - ▶ AI strategy reports
  - ▶ Data + AI conference talks
  - ▶ LinkedIn Learning AI paths